

MONTAUK, NY, USA

WMO: 725014

Lat: 41.073N

Lon: 71.923W

Elev: 2

StdP: 101.30

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 54780

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.9	(c) -7.5	(d) -19.0	(e) 0.7	(f) -8.6	(g) -17.2	(h) 0.8	(i) -6.5	(j) 11.6	(k) -7.1	(l) 10.8	(m) -5.4	(n) 6.2	(o) 320	(p) 0.448

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.6	29.7	23.0	28.0	22.5	26.9	21.9	24.6	27.6	23.9	26.5	23.1	25.5	2.8	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.7	18.5	26.1	22.9	17.7	25.4	22.4	17.1	24.8	74.3	27.6	71.4	26.5	68.5	25.5	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.4	(b) 8.4	(c) 7.6	(e) -12.3	(f) 32.7	(g) 3.9	(h) 2.1	(i) -15.1	(j) 34.2	(k) -17.4	(l) 35.5	(m) -19.6	(n) 36.7	(o) -22.4	(p) 38.3		
(a) 9.4	(b) 8.4	(c) 7.6	(e) -13.5	(f) 25.8	(g) 3.2	(h) 1.0	(i) -15.8	(j) 26.5	(k) -17.7	(l) 27.1	(m) -19.5	(n) 27.6	(o) -21.9	(p) 28.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.8	1.1	1.2	4.1	9.1	13.9	19.1	22.9	22.9	19.6	14.2	9.0	4.4
	DBStd	8.57	4.96	4.25	3.87	3.26	3.17	3.05	2.39	2.47	2.75	3.62	3.96	4.30
	HDD10.0	1018	277	248	187	53	3	0	0	0	0	7	64	179
	HDD18.3	2771	535	481	443	277	142	26	1	1	19	136	280	431
	CDD10.0	1692	1	1	2	26	124	274	400	399	288	137	34	7
	CDD18.3	403	0	0	0	1	5	51	142	142	56	7	0	0
	CDH23.3	1739	0	0	0	4	19	187	706	692	125	6	0	0
	CDH26.7	324	0	0	0	1	2	35	152	123	11	1	0	0
Wind	WSAvg	3.6	4.5	4.3	4.2	3.7	3.0	2.7	2.6	2.4	2.9	3.7	4.1	4.5
Precipitation	PrecAvg	1136	88	79	113	101	86	85	82	102	106	99	87	108
	PrecMax	1395	153	141	360	195	195	281	288	209	219	330	142	189
	PrecMin	876	40	40	12	32	27	0	34	39	24	10	21	23
	PrecStd	147	31	30	72	47	37	65	49	48	49	66	35	37
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.7	12.4	15.7	22.0	25.8	30.3	32.3	32.1	28.0	23.8	18.6	16.2
		MCWB	11.6	10.0	11.1	14.1	19.0	22.6	24.4	24.9	22.6	19.2	16.4	14.0
	2%	DB	11.1	10.3	12.5	17.7	22.7	27.4	30.0	29.3	26.1	22.0	17.2	14.1
		MCWB	9.9	8.8	9.1	12.3	17.2	21.3	23.2	23.2	21.3	19.7	14.9	12.7
	5%	DB	9.7	9.0	10.9	15.4	20.9	25.5	28.2	27.8	24.8	20.6	16.0	12.8
		MCWB	7.9	7.0	8.4	10.7	16.3	19.9	22.7	22.7	20.9	18.0	13.9	11.5
	10%	DB	8.0	7.4	9.3	13.6	18.9	23.8	26.8	26.8	23.6	19.2	14.9	11.2
		MCWB	6.2	5.3	7.0	9.9	15.0	19.1	21.9	22.2	20.4	17.0	12.8	9.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.9	11.3	12.4	15.5	20.3	23.7	25.5	25.8	24.2	21.2	17.3	14.7
		MCDB	12.4	11.6	14.1	19.6	24.2	28.8	30.4	29.8	26.7	22.5	18.3	15.6
	2%	WB	10.2	9.1	10.3	13.6	18.6	22.1	24.5	24.8	23.0	20.0	15.5	13.1
		MCDB	10.9	10.1	11.6	16.1	21.5	25.9	28.1	27.3	24.5	21.3	16.7	13.9
	5%	WB	8.1	7.5	8.8	12.2	17.2	21.1	23.8	24.1	22.0	18.7	14.4	11.6
		MCDB	9.2	8.8	10.3	14.2	19.9	24.0	26.6	26.5	23.6	20.1	15.6	12.7
	10%	WB	6.6	5.6	7.4	10.9	15.8	20.2	23.0	23.4	21.2	17.6	13.3	9.8
		MCDB	7.8	7.2	9.1	12.8	18.1	22.7	25.5	25.6	22.8	19.0	14.6	11.1
Mean Daily Temperature Range	5% DB	MDBR	6.0	6.2	6.4	6.9	7.3	7.0	6.8	6.6	6.3	5.9	6.2	5.8
		MCDBR	7.6	7.8	8.1	9.7	10.0	9.3	8.5	7.6	7.0	6.6	6.9	7.1
		MCWBR	7.8	7.1	6.1	5.5	5.6	4.7	3.9	3.5	4.3	5.2	6.4	7.7
	5% WB	MCDBR	7.4	8.0	7.7	8.0	8.9	7.8	7.2	6.0	5.7	5.6	6.2	7.0
		MCWBR	7.9	8.0	6.8	5.7	5.5	4.6	3.9	3.4	4.2	5.4	6.7	7.9
Clear-Sky Solar Irradiance	taub	0.315	0.323	0.343	0.379	0.428	0.466	0.488	0.454	0.385	0.353	0.322	0.312	
	taud	2.498	2.470	2.448	2.382	2.278	2.216	2.160	2.254	2.414	2.508	2.548	2.547	
	Ebn at Noon	844	891	908	893	852	817	794	812	852	844	830	820	
	Edh at Noon	75	89	101	116	132	141	148	131	104	85	71	67	
All-Sky Solar Radiation	RadAvg	1.83	2.73	3.79	4.92	5.48	5.99	6.08	5.47	4.46	3.02	2.05	1.54	
	RadStd	0.15	0.27	0.30	0.52	0.53	0.59	0.33	0.32	0.34	0.28	0.16	0.14	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	+1.15	+1.25	N/A	N/A	N/A	N/A
(47) Regional (71 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+104	+56

Nomenclature: See separate page